Set

1. **If A={1,2,3} and B={2,3,4}, what is A∪B?**
a. {1,2,3} b.{2,3} c. {1,2,3,4} d. {4}
2. **If A={1,3,5} and B={1,2,3,4,5}, what is A⊆B?**
a. True b. False
3. **What is the Cartesian product A× B if A={1,2} and B={x,y}?**
a. {(1,x),(1,y),(2,x),(2,y)} b. {(1,x),(2,x)}
c. {(x,1),(y,2)} d. None of these
4. **If A ∪ B = A and A ∩ B = A, what can be concluded about A and BB?**
a. A ⊆ B b. B ⊆ A c. A=B d. None of these
5. **If f (x) = x + 3, what is f (5) ?**
a. 8 b. 15 c. 3 d. None
6. **If A={1,2,3} and B={4,5,6}, what is A ∩ B?**
a. {1,2,3} b. {4,5,6} c. {1,4} d. ϕ
7. **If $f(x) = x^2$, is $f(x)$ a one-to-one function?**
a. Yes b. No
8. **What is the domain of $f(x) = \frac{1}{x-2}$?**
a. R b. R/ {2} c. {2} d. None
9. **If $f: R \to R$ is defined by $f(x) = 3x + 2$, find $f(0)$.**
a. 0 b. 2 c. 3 d. -2
10. **The range of $f(x) = x^2$ for $x \in R$ is:**
a. [0, ∞) b. (-∞, 0] c. R d. [0, ∞)
11. **If $f(x) = x + 1$ and $g(x) = x - 1$, what is $f(g(x))$?**
a. X b. x+2 c. x - 1 d. x+1
12. **Which of the following sets is finite?**
a. N b. {1, 2, 3, 4, 5} c. Z d. R
13. **If $A = \{x \mid x > 5\}$, what is the range of possible values for x in A?**
a. [6, ∞) b. (5, ∞) c. {5} d. (-∞,5)
14. **Which of the following is the identity element for addition in real numbers?**
a. 1 b. -1 c. 0 d. None
15. **If $f(x) = x^2$ and $g(x) = x + 1$, then $(f \circ g)(x)$ is:**
a. $x^2 + 1$ b. $(x + 1)^2$ c. $x^2 + 2x + 1$ d. None
16. **What is the power set of {1,2}?**
a. Ø} b. {{1},{2}} c. {Ø , {1}, {2}, {1, 2} } d. None
17. **What is the domain of $f(x) = \sqrt{\{x - 3\}}$?**
a. [3,∞) b. (3, ∞) c. (-∞, 3] d. (-∞, 3)
18. **Which of the following represents a function?**
a. {(1,2),(1,3)} b. {(1,2),(2,3)} c. {(1,1),(1,1)} d. None
19. **The relation $R = \{(1, 1), (2, 2), (3, 3)\}$ on the set {1,2,3}is:**
a. Reflexive b. Symmetric c. Transitive d. All of these
20. **What is the inverse function of $f(x) = x + 3$?**
a. $f^{-1}(x) = x - 3$ b. $f^{-1}(x) = x + 3$ c. $f^{-1}(x) = -x + 3$ d. None

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21. If $A \cap A = \phi$, then $A - A$ is

a. $\{0\}$ b. $\{\phi\}$ c. A d. $\{A\}$

22. An empty set is

a. a set having no subject b. a set having no super set

c. a set having no element d. a set having only one element zero

23. A universal set is

a. the superset of every set under consideration

b. the subset of every set under consideration

c. the set of all real number

d. the set of complex numbers

24. If A={1,3,5,7} and B = {2,4,5,6,7} then $A - \{A \cap B\} =$

a. {1,3,5,7} b. {3, 4} c. {1, 3} d. {5, 6 }

25. If A={1,2,3,4} and B = {3,4,5,6} then $\{A \Delta B\} =$

a. {1,2 ,3,4,5,6} b. {3,4} c. {1, 2 } d. {1, 2, 5, 6}

Hints

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Percentage

1. What is 20% of 200?
a. 20 b. 30 c. 40 d. 50

2. Convert 75% to a fraction.
a. ¼ b. ¾ c. 1/ 2 d. 7/8

3. If 15% of a number is 45, find the number.
a. 200 b. 250 c. 300 d. 350

4. A jacket costs NPR 400 after a 20% discount. What was the original price?
a. NPR 450 b. NPR 500 c. NPR 480 d. NPR 520

5. Express 0.25 as a percentage.
a. 10% b. 20% c. 25% d. 30%

6. If a student scores 72 out of 80, what is the percentage score?
a. 80% b. 85% c. 90% d. 95%

7. What is the percentage increase when a number rises from 50 to 65?
a. 25% b. 30% c. 20% d. 15%

8. 10% of what number is 15?
a. 120 b. 150 c. 180 d. 200

9. If a product's price increases by 25%, what is the new price of an item originally costing NPR 800?
a. NPR 850 b. NPR 1000 c. NPR 950 d. NPR 1200

10. A man spends 60% of his salary and saves NPR 8000. What is his total salary?
a. NPR 20000 b. NPR 24000 c. NPR 15000 d. NPR 30000

11. The population of a town increases by 10% every year. If the current population is 10000, what will it be next year?
a. 11000 b. 12000 c. 12500 d. 10500

12. A student gets 40% marks and fails by 20 marks. The passing mark is 50%. Find the total marks.
a. 100 b. 150 c. 200 d. 250

13. Convert 7/8 into a percentage.
a. 75% b. 85% c. 87.5% d. 90%

14. Find 15% of 120.
a. 150 b. 160 c. 180 d. 200

15. By selling an item for NPR 1200, a shopkeeper gains 20%. What was the cost price?
a. NPR 1000 b. NPR 1100 c. NPR 1050 d. NPR 1150

16. Increase 400 by 25%.
a. 425 b. 450 c. 475 d. 500

17. Decrease 800 by 10%.
a. 720 b. 760 c. 700 d. 750

18. Find 250% of 60.
a. 120 b. 150 c. 180 d. 200

19. A quantity is reduced by 40% and then increased by 50%. What is the overall percentage change?
a. 5% increase b. 10% increase c.5% decrease d. No change

20. A student scored 36 out of 50 marks. What is the percentage?
a. 60% b. 70% c. 72% d. 75%

1. C	**2. B**	**3. C**	**4. B**	**5. C**	**6. C**	**7. B**	**8. B**	**9. B**	**10. A**

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